

Student Management System - JavaScript Project Requirements

This project is designed to help practice JavaScript fundamentals including DOM manipulation, arrays, objects, functions, loops, conditions, sorting, searching, and CRUD operations.

Project Objective

Build a Student Management System where users can add, update, delete, search, sort, and display student records using JavaScript.

Required Features

1. Add Student
2. Update Student
3. Delete Student
4. Search Student by Name
5. Display All Students
6. Sort Students by Marks (Ascending & Descending)
7. Show Pass/Fail Result
8. Display Total Student Count

Student Object Structure

```
{ id: 1, name: "Raj", age: 20, marks: 85 }
```

Topics That Must Be Used

DOM Manipulation (querySelector, innerHTML)
Variables (let, const)
Data Types
Arrays & Objects
Functions
Conditions (if-else, switch if needed)
Loops (for-of, for-in)
Array Methods (push, filter, map, find, sort)
Template Literals
Optional Chaining (bonus)

Suggested UI

Student Form:

- Name Input
- Age Input
- Marks Input
- Add Button

Student Table:

- ID
- Name

- Age
- Marks
- Result
- Actions (Edit/Delete)

Functions to Implement

addStudent()
displayStudents()
updateStudent(id)
deleteStudent(id)
searchStudent(name)
sortStudents()
getResult(marks)

Bonus Features

Local Storage
Dark Mode
Validation
Export Data
Statistics Section

Evaluation Checklist

- CRUD Operations Working
- Search Working
- Sorting Working
- Pass/Fail Logic Working
- Clean UI
- Proper Use of Arrays and Objects
- Code Comments Added